

Answers

VARC

1.C	2.D	3.A	4.B	5.C	6.D	7.A	8.B
9.D	10.A	11.C	12.B	13.B	14.D	15.C	16.A
17.B	18.2413	19.1	20.5	21.3214	22.D	23.2143	24.A

LRDI

25.6	26.11	27.B	28.C	29.C	30.C	31.12	32.D
33.B	34.A	35.A	36.D	37.B	38.D	39.5	40.C
41.6	42.B	43.B	44.A				

Quant

45.7	46.C	47.4195	48.D	49.1000	50.A	51.B	52.45
53.A	54.5	55.A	56.D	57.D	58.C	59.C	60.250
61.10	62.B	63.D	64.30	65.A	66.A		

Explanations

VARC

1. C

The argument in Option A has been discussed in the following excerpt:

...the sceptic's contention that we cannot be sure that we know anything - at least not if we think of knowledge as something like having a correct mental representation of reality, and not if we think of reality as something like things-as-they-are-in-themselves, independent of our perceptions, ideas or descriptions.

The argument in Option B has been discussed in the following excerpt:

For, they explain, unless we can ground our claims to knowledge as such, which is to say, distinguish it from mere opinion, superstition, fantasy, wishful thinking, ideology, illusion or delusion, then the actions we take on the basis of presumed knowledge - boarding an airplane, swallowing a pill, finding someone guilty of a crime - will be irrational and unjustifiable.

The argument in Option D has been discussed in the following excerpt:

While it is not clear that the scandal matters to anyone but philosophers, philosophers point out that it should matter to everyone, at least given a certain conception of knowledge.

The author does say that ignoring the scepticism about the veracity of knowledge is the most common way of dealing with it, not the best way. Hence. Option C has not been discussed.

VIDEO SOLUTION

2. D

...and not if we think of reality as something like things-as-they-are-in-themselves, **independent of our perceptions, ideas or descriptions**. For, the sceptic will note, since **reality, under that conception of it, is outside our ken** (we cannot catch a glimpse of things-in-themselves around the corner of our own eyes; we cannot form an idea of reality that floats above the processes of our conceiving it), we have no way to compare our mental representations with things-as-they-are-in-themselves and therefore no way to determine whether they are correct or incorrect.

The author is making a logical argument in the sentence given in the question. According to the author, if we say that the reality of objects is independent of our perceptions, then it is out of our ken. Hence, we cannot for this idea of reality, and logically, we would be unable to comprehend it. Option D comes the closest to capturing this point.

Option A is incorrect. The sceptic argues that if reality were independent of individual perceptions, then the reality would float above the processes with which we conceive it. It is being presented as an argument to negate the viewpoint, while the Option takes it as an established fact.

Perception is more than eyesight. Option B captures only eyesight, and hence, is a distortion.

The process of conceiving reality and our knowledge is not the argument the author presents in the mentioned lines. Hence, Option C is incorrect.

VIDEO SOLUTION

3. A

We hold our beliefs and presumptive knowledges more or less confidently, usually **depending on how we acquired them** (I saw it with my own eyes; I heard it on Fox News; a guy at the office told me) **and how broadly and strenuously they seem to be shared or endorsed by various relevant people**: experts and authorities, friends and family members, colleagues and associates.

From the above excerpt, it is clear that held beliefs and presumptive knowledges depend upon how we acquired them, and how strongly they are shared or endorsed by relevant people in our circle. Hence, Option A is the answer.

VIDEO SOLUTION

4. B

We hold our beliefs and presumptive knowledges more or less confidently, usually depending on how we acquired them...

As mentioned in the above line, Option A directly contradicts what the author says.

It has been said that knowledge, or the problem of knowledge, is the scandal of philosophy. The scandal is philosophy's apparent inability to show how, when and why we can be sure that we know something or, indeed, that we know anything.

The author then goes on to explain that sceptic view is the skeleton in western philosophy's closet when trying to negate this scandal. Thus, it means that the scandal has to do with the sceptic way of thinking.

In the next paragraph, the sceptic view has been explained, which talks about our inability to grasp reality if we think of it as independent of our perceptions. Hence, the scandal can be construed to be the same. The author is likely to agree with this view. Option B is the answer.

In the last paragraph, the author points out that we defend a viewpoint strongly if we feel that it is held widely in our social circle. However, the author does not allude to the fact that it is appropriate or not. Also, such an argument would bolster the inherent bias we have, and hence would point towards a fault in our decision making. Hence, Option C is incorrect.

Option D is not supported in the passage. The passage presents the sceptic view that if reality were construed as independent of our perceptions, then it would be impossible to grasp reality. It has not been mentioned that in this case, we should aim to study that reality in a similar manner.

VIDEO SOLUTION

5. C

Through war, famine and natural disasters, whole communities can be destroyed, taking their language with them to the grave...

Such trading up and out of a speech form occurs for complex political, cultural and economic reasons - sometimes voluntary for economic and educational reasons, although often amplified by state coercion or neglect.

More commonly, speakers live on but abandon their language in favor of another vernacular, a widespread process that linguists refer to as "language shift" from which few languages

From the above excerpts, Options A, B, and D can be supported. The passage supports that Internet technology has allowed certain endangered languages to thrive. Option C is not supported anywhere in the passage, hence, is the answer.

VIDEO SOLUTION



6. D

It's easy to forget that most of the world's languages are still transmitted orally with no widely established written form. While speech communities are increasingly involved in projects to protect their languages - in print, on air and online - orality is fragile and contributes to linguistic vulnerability. But indigenous languages are about much more than unusual words and intriguing grammar: They function as vehicles for the transmission of cultural traditions, environmental understandings and knowledge about medicinal plants, all at risk when elders die and livelihoods are disrupted.

From the underlined portions of the above excerpt, we can infer options A, B, and C.

The first paragraph does mention that indigenous languages have unusual words and intriguing grammar. But it has not been mentioned as a factor that makes their documentation challenging. Hence, Option D is the answer.

VIDEO SOLUTION

7. A

Multilingualism can help us live in a more connected and more interdependent world.

Throughout the passage, the author is advocating for greater language diversity and suggesting how to counter the threat to the same. Multilingualism is something that the author has supported as cited above. Hence, Option A is the answer.

More commonly, speakers live on but abandon their language in favor of another vernacular, a widespread process that linguists refer to as “language shift” from which few languages are immune.

Since the author advocated the preservation of languages, he would likely be against this shift, as it endangers languages. Option B can be eliminated.

Again, the author is pushing for more diversity and preservation of cultures instead of homogenization of the same. Option C can be eliminated.

The author does not push for state intervention in the preservation of languages. Hence, Option D would not be the answer.

 VIDEO SOLUTION

8. B

More commonly, speakers live on but abandon their language in favor of another vernacular, a widespread process that linguists refer to as “language shift” from which few languages are immune. Such trading up and out of a speech form occurs for complex political, cultural and economic reasons - sometimes voluntary for economic and educational reasons, although often amplified by state coercion or neglect. Welsh, long stigmatized and disparaged by the British state, has rebounded with vigor.

In the above excerpt, the trading of language for another (language shift) has been mentioned. The author mentions the reasons why this happens, and then the Welsh language is mentioned as an example which has rebounded against the same with vigour. Thus, it is presented as a ray of hope, that a language can be revived even when cultural shift occurs. Option B is the answer.

The example has not been mentioned to spite the efforts that were put to integrate Welsh speakers into English speaking fold. The purpose of the author is to deal with endangered languages, and the subject in Option A would be out of scope.

The role of state effort in revitalising Welsh has not been mentioned, hence, Option C can be eliminated.

The role of globalisation in revitalising Welsh has not been mentioned either. Hence, Option D can be eliminated too.

 VIDEO SOLUTION

9. D

I have elaborated . . . a framework for analyzing the contradictory pulls on [Indian] nationalist ideology in its struggle against the dominance of colonialism and the resolution it offered to those contradictions. Briefly, this resolution was built around a separation of the domain of culture into two spheres—the material and the spiritual. It was in the material sphere that the claims of Western civilization were the most powerful. Science, technology, rational forms of economic organization, modern methods of statecraft—these had given the European countries the strength to subjugate the non-European people . . . To overcome this domination, the colonized people had to learn those superior techniques of organizing material life and incorporate them within their own cultures. . . . But this could not mean the imitation of the West in every aspect of life, for then the very distinction between the West and the East would vanish—the self-identity of national culture would itself be threatened. . . .

The first paragraph acknowledges that the nationalist ideology was fighting against colonial dominance, and there were certain inherent contradictions in the way this struggle was being carried out. The author says that a method of resolution of these contradictions was to separate material and spiritual domains. This hints at the contradiction present. We can infer from here that this is being done because the nationalists acknowledge that the colonial countries were superior in certain aspects, which allowed them to subjugate non-Europeans, as is also mentioned later in the paragraph. The author also mentions that the colonized people had to learn those superior techniques, instead of all-out rejection of colonialist ideas and the progress they brought with them. Thus, the contradiction was that to overcome colonial dominance, nationalism had to accept that the material ways of the West were superior and incorporate them. Hence, Option D is the answer.

 VIDEO SOLUTION

10. A

The discourse of nationalism shows that the material/spiritual distinction was condensed into an analogous, but ideologically far more powerful, dichotomy: that between the outer and the inner...

The above excerpt shows that the material/spiritual distinction was condensed to form a far more superior dichotomy of the outer and the inner. Thus, the former was the premise for the latter, as well as inferior to the latter. Hence, Options B and C are true.

To overcome this domination, the colonized people had to learn those superior techniques of organizing material life and incorporate them within their own cultures. . . . But this could not mean the imitation of the West in every aspect of life, for then the very distinction between the West and the East would vanish—the self-identity of national culture would itself be threatened. . . .

From the above excerpt, we can infer that the dichotomy helped save the identity of Indian Nationalism. Option D is also true.

Option A is not true as per the passage, and hence, is the answer.

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11. C

The arguments in the passage are based on the premise that the material and spiritual aspects of culture were different. Hence, even if Indian nationalism accepted the superior material ways of the west, they still would not be giving in to colonial dominance and their identity would be preserved by the spiritual aspect, as it remained unaffected. Hence, to weaken the author's argument, we can give a statement that proves that the spiritual aspect was affected too. Option C does that and is the answer.

Rejecting education for women could have more than one reason. It does not imply that the spiritual part of Indian culture was affected by colonialism. Moreover, the separation of roles according to gender is something that nationalist ideology supported, hence, rejecting education based on gender would not contradict it. Option A can be eliminated.

Option B does not weaken the author's argument since the author already agrees that the forces of colonialist modernity helped shape Indian nationalism, but only in the material aspect.

Option D is unrelated to the argument at hand and can be eliminated too.

VIDEO SOLUTION

12. B

Once we match this new meaning of the home/world dichotomy with the identification of social roles by gender, we get the ideological framework within which nationalism answered the women's question. It would be a grave error to see in this, as liberals are apt to in their despair at the many marks of social conservatism in nationalist practice, a total rejection of the West. Quite the contrary: the nationalist paradigm in fact supplied an ideological principle of selection.

From the above excerpt, we can see that the liberals were concerned over the social conservatism that nationalist practice promoted as an ideological principal of selection, where social roles would be selected according to the gender of the person. Hence, Option B is the answer, as it comes the closest in capturing the liberal perception of the same.

The material/spiritual dichotomy has not been discussed in terms of liberal perspective, hence, Option A is out of the score here.

Option C is incorrect. Indian nationalism did not accept the changes brought about the colonialism, rather, promoted the segregation of gender roles according to their spiritual ideology of home/ world dichotomy.

Option D is contrary to what is mentioned in the passage. The author says that the 'ideological principle of selection' was the actual truth, and the liberal perspective was just contrary to what was actually happening.

VIDEO SOLUTION

13. B

The author begins that passage by saying that truth does not necessarily carry power. He then goes on to explain that to attain social cohesion, sticking to the truth is not always an optimal strategy. In the last paragraph, the author sums up this trade-off:

Even if we need to pay some price for deactivating our rational faculties, the advantages of increased social cohesion are often so big that fictional stories routinely triumph over the truth in human history. Scholars have known this for thousands of years, which is why scholars often had to decide whether they served the truth or social harmony.

Thus, Option B is the answer.

The author is not fixated upon the types of stories, not upon what kind of stories do the leaders propagate. Hence, Options A and C can be eliminated.

Power has been mentioned to indicate that sometimes, absolute truth is not the way forward to achieve maximum utility. The main contention of the author is not the trade-off between truth and power but between truth and social cohesion. Option D can be eliminated too.

VIDEO SOLUTION

14. D

The author does not support that there is a limit to the influence that myths have on people, nor does he support imposing one. Hence, Option A can be eliminated.

Third, and most important, the truth is often painful and disturbing. Hence if you stick to unalloyed reality, few people will follow you. An American presidential candidate who tells the American public the truth, the whole truth and nothing but the truth about American history has a 100 percent guarantee of losing the elections. . . . An uncompromising adherence to the truth is an admirable spiritual practice, but it is not a winning political strategy. . . .

Option B is contrary to what is being said in the passage. The author says that untarnished truth is not a good recipe for a political win, hence, the candidate should steer clear of that.

Option C is also contrary to what is being said in the passage. According to the author, not conveying the complete truth will allow a person to stay in power.

Option D is in line with the above excerpt and hence, is the answer.

VIDEO SOLUTION

15. C

Even if we need to pay some price for deactivating our rational faculties, the advantages of increased social cohesion are often so big that fictional stories routinely triumph over the truth in human history. Scholars have known this for thousands of years, which is why scholars often had to decide whether they served the truth or social harmony. Should they aim to unite people by making sure everyone believes in the same fiction, or should they let people know the truth even at the price of disunity?

In the penultimate paragraph, the author mentions how successful leaders balance truth and social unity to achieve an optimal outcome. The above excerpt shows that scholars have known this for a long time too, and have implemented it. Thus, Option C is the answer.

That leaders and scholars have improved with time when it comes to achieving social cohesion is not implied in the passage. Option A can be eliminated.

We cannot say that scholars use myths to obtain power as leaders do. Hence, Option B can be eliminated.

We cannot say that scholars use myths to stay in office as leaders do. Option D can be eliminated too.

VIDEO SOLUTION



16. A

Option A: The author here emphasizes that the choice between truth and social cohesion is a difficult one for scholars, as it means choosing between truth or uniting everyone using a common narrative. Here, the reach and influence of fiction created by that scholar has been overemphasized, and hence, is the answer.

Option B: There is no overemphasis in this option. Since humans have achieved these feats, and these feats do manipulate the objective reality around us, Option B can be eliminated.

Option C: Here too, the importance of fiction has not been overemphasized, but the importance of truth has been downplayed.

Option D: Option D presents a statement, which is unrelated to the emphasis being placed on the importance of myths.

VIDEO SOLUTION

17. B

The main points of the paragraph are:

1. As compared to physicists, biologists are more conservative when it comes to the subject of preprints.
2. Preprints allow faster dissemination of knowledge.

A: Misses out the comparison between biologists and physicists.

B: Captures both the points appropriately and is the answer.

C: Also misses out the comparison between biologists and physicists.

D: Factually incorrect, physicists and not biologists are open to the idea of preprints.

VIDEO SOLUTION

18. 2413

A brief reading of the sentences suggests that the paragraph is about the change needed in the way we go about development. 2 introduces the topic at hand, that this is a watershed moment when it comes to the subject of development. 41 make a mandatory pair, which talks about what the purpose of development was at the beginning and how it needs to be altered to suit the needs of today. 3 then aptly ends the paragraph, suggesting the measures that could be taken to counter the same. Hence, the proper sequence would be 2413.

VIDEO SOLUTION

19. 1

The sentences have been taken from Harvard's Justice, and have been modified considerably. Since a paragraph has not been directly taken here, the better way of elimination here would be to evaluate the major points of each sentence and see which one runs tangent to the discussion at hand. (During the examination, one must try both ways to solve: arranging and eliminating.)

1. Using the debate on biotechnology to evaluate moral arguments.
2. Why is bioengineering disputed?
3. The promise of bioengineering.
4. Ethics of bioengineering based on theology.
5. The theological argument.

We can see here that the last four sentences try to examine why bioengineering is disputed in spite of its huge potential. Then reasons are given about the question on its ethicality, and how it is closely associated with theology on the matter.

1 however runs tangential to the discussion. The main focus is bioengineering while 1 aims to shift the focus and use the debate on the matter as a stepping stone to reach another goal: evaluating/formulating moral arguments. Hence, 1 is the odd one out here.

VIDEO SOLUTION

20. 5

A brief reading of the sentences tells us that the paragraph must be about the industry of neuromarketing, which is still in its embryonic phase. 3 can be the opening sentence to the paragraph, as it introduces the topic at hand. All the other sentences need a sentence before them that introduces what is being talked about.

1, 4, and 2 then go on to talk about the opinion of different associated parties on the matter. It has taken the advertising industry by storm. Others feel that this 'art' is being masked as a science, and many lack enthusiasm on the matter.

5 however, does not fit in here. The reason is that it talks about 'psychology', which is different from the use of neural science. Even if one is not familiar with the difference, we can see that it goes a step forward to talk about the application of a science, whereas the paragraph is mostly concerned with a growing science and how it is shaping public opinion. Hence, 5 is the odd one out.

VIDEO SOLUTION



21. 3214

A brief reading of the sentences suggests that the paragraph is about great inventions, focusing on the importance of scientific method and how it forms the foundation for other great inventions. 32 forms an introductory pair that claims that the modern scientific method must be the greatest of all inventions and then talks about its mode of inquiry.

1 then presents an invention that could solve many problems in the future. 4 then claims that the invention would have been impossible if the scientific method did not precede it. Hence, the coherent arrangement is 3214.

VIDEO SOLUTION

22. D

The main points of the paragraph are:

1. The value of creativity to economic progress has been realised, with serious investment being done to study/promote it.
2. But this success fires back. Policing its meaning can lead to obsession, hampering creativity itself.

A: It is extreme in approach. The paragraph does not imply that it has become impossible to define the concept, but it becomes difficult to practice creativity when it is being forced on oneself.

B: It is also extreme. The obsession hampers, not completely destroys the creative process.

- C: This option is a distortion and fails to capture the above points.
 D: Comes the closest to capturing the above two points, and hence, is the answer.

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23. 2143

A brief reading of the sentences suggests that the paragraph is about the dispute between the US and Canada over the Northwest passage.

2 is a better opening sentence than 1 as 21 forms a good introduction into the dispute. 2 mentions that Canada claims ownership over the passage, and 1 then mentions the counterclaim the US offers: the passage is an international route.

43 becomes a mandatory pair which follows 21. 4 mentions the historical development in the dispute: it has not been legally settled yet. 3 then hints about the future of this dispute: the dispute itself would vanish once the entire Arctic begins to stay ice-free in summer.

Hence, the coherent arrangement is 2143.

[VIDEO SOLUTION](#)

24. A

The main points of the paragraph are:

1. In an economic disaster, the atypical alliance of established industrialist and unemployed workers proves powerful.
2. Anxieties and anticipation lead them to look after their interests.
3. It is the industrialist that benefits the most as he is able to use the latter to achieve his vested political interests.

A: Comes the closest in capturing all three points, and hence, is the answer.

B: Distortion. This purpose has not been mentioned in the passage.

C: Distortion. It has not been mentioned that an economic downturn creates competition. It has been mentioned that during such a disaster, the industrialist is more likely to focus on the downsides emerging from free-market (competition) than the upsides.

D: Distortion. It has been implied that the industrialist manipulates the situation to fulfill his own political agenda, and not that the two parties come together to achieve a single goal.

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LRDI

25. 6

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now No booking days from the table are : September 8,9,10,11,12 and 14.

So a total of 6 days .

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26. 11

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now Envelopes and Accessories has only 1 ship mode i.e Standard class .

So therefore processing days for envelopes = 7-3 =4

and processing days for accessories = 19-1 =18

Therefore average = $\frac{(18+4)}{2} = 11$

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27.B

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now taking average processing time per order for the above mentioned categories we get :

$$\text{Art} = \frac{2+8+2+1+7}{5} = 4$$

$$\text{Binders} = \frac{1+1+11+2}{4} = 3.75$$

$$\text{Papers} = \frac{3+2+5}{3} = 3.33$$

$$\text{Phones} = \frac{2+12+1}{3} = 5$$

So in decreasing order we get Phones , Art ,Binder , Paper.

[VIDEO SOLUTION](#)

28.C

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now from the table we observe that the total number of orders are 35 and 7 orders have a processing time of 1 unit

The 7 orders are : Arts Standard class, Binders First class and standard class, Phones First class, Bookcases second class (2 orders) and Chairs standard class.

So the percentage = $\frac{7}{35} \times 100 = 20$

[VIDEO SOLUTION](#)

Explanation [29 - 34]:

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = 8+8

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = 9+7

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with O4 and o6

So if Disha receives o4 and o6

The valuation of Disha will be 5+3 =8 which is not an odd number

so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = 12 =9+3.

Now we can say Charu will have 16 as his own valuation so he will get 8+8 .

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

29. C

So o8 is given to Charu.

[VIDEO SOLUTION](#)

30. C

So Amar envies someone else

[VIDEO SOLUTION](#)

31.12

Amar's own valuation = $9+3=12$

[VIDEO SOLUTION](#)

32.D

o4 is given to Disha

[VIDEO SOLUTION](#)

33.B

o5 is given to Elise

[VIDEO SOLUTION](#)

34.A

So o1 is given to Disha

[VIDEO SOLUTION](#)

Explanation [35 - 38]:

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

Group - A	Group - B
Aruna	vs Biju
Azul vs Aruna	Biju vs Brinda/ Brij
Azul vs Arif	Biju vs Brinda/ Brij

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

Group - C	Group - D
Chitra	vs Dipen
Chetan vs Chitra	Dipen vs Deb/Donna
Chetan vs Chavvi	Dipen vs Deb/Donna

Aruna plays Chitra in the finals and wins the final round.

35.A

Dipen was ranked 2 or 3 in his group

[VIDEO SOLUTION](#)

36.D

Chitra and Dipen played in the semifinals

37. **B**

Aruna is the winner.

38. **D**

Brij was the player from group B who played Chitra. Aruna played in finals, Chetan in round 2, and Dipen in semi finals

Explanation [39 - 44]:

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

39.5

The minimum rating awarded is 5.

[VIDEO SOLUTION](#)

40. C

Bihari and Chirag has given the highest ratings

[VIDEO SOLUTION](#)



41. 6

Atal has given a rating of 6 in timeliness

[VIDEO SOLUTION](#)

42. B

Among Atal and Deepak, one of them gives a tip of 30 and the other gives a tip of 50. Hence 30 or 50 any case is possible

[VIDEO SOLUTION](#)

43. B

Atal has given the maximum rating in Behaviour.

[VIDEO SOLUTION](#)

44. A

Deepak gives a rating of 7 in Packaging.

[VIDEO SOLUTION](#)

Quant

45. 7

$$2.25 \leq 2 + 2^{n+2} \leq 202$$

$$2.25 - 2 \leq 2 + 2^{n+2} - 2 \leq 202 - 2$$

$$0.25 \leq 2^{n+2} \leq 200$$

$$\log_2 0.25 \leq n + 2 \leq \log_2 200$$

$$-2 \leq n + 2 \leq 7.x$$

$$-4 \leq n \leq 7.x - 2$$

$$-4 \leq n \leq 5.x$$

Possible integers = -4, -3, -2, -1, 0, 1, 2, 3, 4, 5

If we see the second expression that is provided, i.e

$3 + 3^{n+1}$, it can be implied that n should be at least -1 for this expression to be an integer.

So, n = -1, 0, 1, 2, 3, 4, 5.

Hence, there are a total of 7 values.

[VIDEO SOLUTION](#)



46. C

Given x, y, z are three terms in an arithmetic progression.

Considering $x = a, y = a+d, z = a+2d$.

Using the given equation $x \cdot y \cdot z = 5 \cdot (x+y+z)$

$$a(a+d)(a+2d) = 5(a+a+d+a+2d)$$

$$= a(a+d)(a+2d) = 5(3a+3d) = 15(a+d).$$

$$= a(a+2d) = 15.$$

Since all x, y, z are positive integers and $y-x > 2$, $a, a+d, a+2d$ are integers.

The common difference is positive and greater than 2.

Among the different possibilities are : $(a=1, a+2d=5), (a=3, a+2d=5), (a=5, a+2d=3), (a=15, a+2d=1)$

Hence the only possible case satisfying the condition is :

$$a = 1, a+2d = 15.$$

$$x = 1, z = 15.$$

$$z-x = 14.$$

[VIDEO SOLUTION](#)

47. 4195

Given the 4 digit number :

Considering the number in thousands digit is a , number in the hundredth digit is b , number in tens digit is c , number in the units digit is d .

Let the number be $abcd$.

$$\text{Given that } a+b+c = 14. \quad (1)$$

$$b+c+d = 15. \quad (2)$$

$$c = d+4. \quad (3).$$

In order to find the maximum number which satisfies the condition, we need to have $abcd$ such that a is maximum which is the digit in thousands place in order to maximize the value of the number. b, c , and d are less than 9 each as they are single-digit numbers.

$$\text{Substituting (3) in (2) we have } b+d+4+d = 15, b+2d = 11. \quad (4)$$

$$\text{Subtracting (2) and (1) : } (2) - (1) : d = a+1. \quad (5)$$

Since c cannot be greater than 9 considering c to be the maximum value 9 the value of d is 5.

$$\text{If } d = 5, \text{ using } d = a+1, a = 4.$$

Hence the maximum value of $a = 4$ when $c = 9, d = 5$.

$$\text{Substituting } b+2d = 11. b = 1.$$

The highest four-digit number satisfying the condition is 4195

[VIDEO SOLUTION](#)

48. D

Raj invested Rs 10000 in the first year. Assuming the loss he faced was $x\%$.

$$\text{The amount after 1 year is } 10,000 \cdot (1 - x/100) = 10000 - 100x.$$

Given the balance was greater than Rs 5000 and hence $x < 50$ percent.

When Raj invested this amount in the second year he earned a profit which is five times that of the first-year percentage.

$$\text{Hence the amount after the second year is : } (10000 - 100x)(1 + \frac{5x}{100}).$$

Raj gained a total of 35 percent over the period of two years and hence the 35 percent is Rs 3500.

Hence the final amount is Rs 13,500.

$$(10000 - 100x)(1 + \frac{5x}{100}) = 13,500$$

$$(100 + 5 \cdot x) \cdot (100 - x) = 13500$$

$$10000 - 100x + 500x - 5x^2 = 13500.$$

$$5x^2 - 400x + 3500 = 0$$

Solving the equation the roots are :

$$x = 10, x = 70.$$

Since $x < 50$, $x = 10$ percent.

[VIDEO SOLUTION](#)

49. 1000

This question is an application of the product rule in probability and combinatorics.

In the product rule, if two events A and B can occur in x and y ways, and for an event E , both events A and B need to take place, the number of ways that E can occur is xy . This can be expanded to 3 or more events as well.

Event 1: Distribution of balloons

Since each child gets at least 4 balloons, we will initially allocate these 4 balloons to each of them.

So we are left with $15 - 4 \times 3 = 15 - 12 = 3$ balloons and 3 children.

Now we need to distribute 3 identical balloons to 3 children.

This can be done in ${}^{n+r-1}C_{r-1}$ ways, where $n = 3$ and $r = 3$.

So, number of ways $= {}^{3+3-1}C_{3-1} = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$

Event 2: Distribution of pencils

Since each child gets at least one pencil, we will allocate 1 pencil to each child. We are now left with $6 - 3 = 3$ pencils.

We now need to distribute 3 identical pencils to 3 children.

This can be done in ${}^{n+r-1}C_{r-1}$ ways, where $n = 3$ and $r = 3$.

So, number of ways $= {}^{3+3-1}C_{3-1} = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$

Event 3: Distribution of erasers

We need to distribute 3 identical erasers to 3 children.

This can be done in ${}^{n+r-1}C_{r-1}$ ways, where $n = 3$ and $r = 3$.

So, number of ways $= {}^{3+3-1}C_{3-1} = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$

Applying the product rule, we get the total number of ways = $10 \times 10 \times 10 = 1000$.

 VIDEO SOLUTION

50. A

Considering the length of train A = L_a , length of train B = L_b .

The speed of train A be $5x$, speed of train B be $3x$.

From the information provided :

The front end of A crossed the rear end of B 46 seconds after the front ends of the trains had crossed each other.

In this case, train A traveled a distance equivalent to the length of train B which is L_b at a speed of $5x + 3x = 8x$ because both the trains are traveling in the opposite direction.

Hence $(8x) \cdot (46) = L_b$.

In the information provided :

It took another 69 seconds for the rear ends of the trains to cross each other.

In the next 69 seconds

The train B traveled a distance equivalent to the length of train A in this 69 seconds.

Hence $(8x) \cdot (69) = L_a$.

$L_a/L_b = 69/46 = 3/2 = 3 : 2$

 VIDEO SOLUTION

100+ CAT Quant Questions

**51. B**

Given a, b, c are rational numbers.

Hence a, b, c are three numbers that can be written in the form of p/q .

Hence if one both the root is $2 + \sqrt{3}$ and considering the other root to be x .

The sum of the roots and the product of the two roots must be rational numbers.

For this to happen the other root must be the conjugate of $2 + \sqrt{3}$ so the sum and the product of the roots are rational numbers which are represented by:

$$-\frac{b}{a}, \frac{c}{a}$$

Since the quadratic equation has negative b , it will cancel the negative sign of the sum of the roots. Hence, the sum of the roots and the products of the roots will be represented by $-\frac{b}{a}, \frac{c}{a}$

Hence the sum of the roots is $2 + \sqrt{3} + 2 - \sqrt{3} = 4$.

The product of the roots is $(2 + \sqrt{3}) \cdot (2 - \sqrt{3}) = 1$

$b/a = 4, c/a = 1$.

$b = 4a, c = a$.

Since $b = c^3$

$4a = a^3$

$a^2 = 4$.

$a = 2$ or -2 .

$|a| = 2$

 VIDEO SOLUTION

52. 45

Let initial volume be V , final be F for milk.

The formula is given by : $F = V \cdot \left(1 - \frac{K}{V}\right)^n$ n is the number of times the milk is drawn and replaced.

so we get $F = V \left(1 - \frac{K}{V}\right)^2$

here $K = 9$

we get

$$\frac{16}{25}V = V \left(1 - \frac{9}{V}\right)^2$$

$$\text{we get } 1 - \frac{9}{V} = \frac{4}{5} \text{ or } -\frac{4}{5}$$

$$\text{If considering } 1 - \frac{9}{V} = -\frac{4}{5}$$

$V = 5$, but this is not possible because 9 liters is drawn every time.

$$\text{Hence : } 1 - \frac{9}{V} = \frac{4}{5}, V = 45 \text{ liters}$$

VIDEO SOLUTION

53. A

All the sides of the rhombus are equal.

The area of a rhombus is 12 cm^2

Considering d_1 to be the length of the longer diagonal, d_2 to be the length of the shorter diagonal.

$$\text{The area of a rhombus is } \left(\frac{1}{2}\right) (d_1) \cdot (d_2) = 12$$

$$d_1 \cdot d_2 = 24.$$

The length of the side of a rhombus is given by $\frac{\sqrt{d_1^2 + d_2^2}}{2}$. This is because the two diagonals and a side form a right-angled triangle with sides $d_1/2$, $d_2/2$ and the side length.

$$\frac{\sqrt{d_1^2 + d_2^2}}{2} = 5$$

$$\text{Hence } \sqrt{d_1^2 + d_2^2} = 10$$

$$d_1^2 + d_2^2 = 100$$

$$\text{Using } d_1 \cdot d_2 = 24, 2 \cdot d_1 \cdot d_2 = 48.$$

$$d_1^2 + d_2^2 + 2 \cdot d_1 \cdot d_2 = 100 + 48 = 148$$

$$d_1^2 + d_2^2 - 2 \cdot d_1 \cdot d_2 = 100 - 48 = 52$$

$$d_1 + d_2 = \sqrt{148} \quad (1)$$

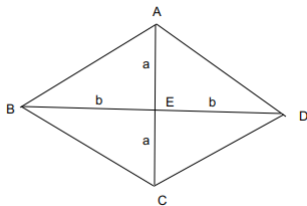
$$d_1 - d_2 = \sqrt{52} \quad (2)$$

$$(1) + (2) = 2 \cdot d_1 = 2 \cdot (\sqrt{37} + \sqrt{13})$$

$$d_1 = \sqrt{37} + \sqrt{13}$$

or

In a rhombus the area of a Rhombus is given by :



The diagonals perpendicularly bisect each other. Considering the length of the diagonal to be $2a$, $2b$.

$$\text{The area of a Rhombus is : } \left(\frac{1}{2}\right) \cdot (2a) \cdot (2b) = 12$$

$$ab = 6.$$

$$\text{The length of each side is : } \sqrt{a^2 + b^2} = 5, a^2 + b^2 = 25,$$

$$(a + b)^2 = 37, (a + b) = \sqrt{37}$$

$$((a - b)^2 = 13, a - b = \sqrt{13})$$

$$2a = (\sqrt{37} + \sqrt{13}), 2b = (\sqrt{37} - \sqrt{13}).$$

$$2a \text{ is longer diagonal which is equal to } (\sqrt{37} + \sqrt{13})$$

VIDEO SOLUTION

54. 5

We have :

$$\log_2 \{3 + \log_3 \{4 + \log_4 (x - 1)\}\} = 2$$

$$\text{we get } 3 + \log_3 \{4 + \log_4 (x - 1)\} = 4$$

$$\text{we get } \log_3 (4 + \log_4 (x - 1)) = 1$$

$$\text{we get } 4 + \log_4 (x - 1) = 3$$

$$\log_4 (x - 1) = -1$$

$$x - 1 = 4^{-1}$$

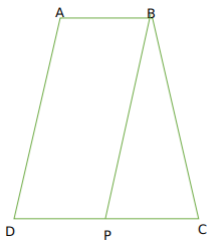
$$x = \frac{1}{4} + 1 = \frac{5}{4}$$

$$4x = 5$$

VIDEO SOLUTION

55. A

We are given that :



Let $DP = x$

So $AB = x$

Now $DP = PC$

So $CD = 2x$

Now let height of trapezium be h

we can say $A(\text{Parallelogram } ABPD) = xh$

And $A(\text{BPC}) = \frac{1}{2}xh$

Now by condition $xh - \frac{1}{2}xh = 10$

$$\frac{xh}{2} = 10$$

so $xh = 20$

Now therefore area of trapezium $ABCD = \frac{1}{2}(x + 2x)h = \frac{3}{2}xh = 30$

VIDEO SOLUTION

100+ CAT DILR Questions



56. D

$$f(x) = \frac{x^2 + 2x + 4}{2x^2 + 4x + 9}$$

If we closely observe the coefficients of the terms in the numerator and denominator, we see that the coefficients of the x^2 and x in the numerators are in ratios 1:2. This gives us a hint that we might need to adjust the numerator to decrease the number of variables.

$$\begin{aligned} f(x) &= \frac{x^2 + 2x + 4}{2x^2 + 4x + 9} = \frac{x^2 + 2x + 4.5 - 0.5}{2x^2 + 4x + 9} \\ &= \frac{x^2 + 2x + 4.5}{2x^2 + 4x + 9} - \frac{0.5}{2x^2 + 4x + 9} \\ &= \frac{1}{2} - \frac{0.5}{2x^2 + 4x + 9} \end{aligned}$$

Now, we only have terms of x in the denominator.

The maximum value of the expression is achieved when the quadratic expression $2x^2 + 4x + 9$ achieves its highest value, that is infinity.

In that case, the second term becomes zero and the expression becomes $1/2$. However, at infinity, there is always an open bracket ')

To obtain the minimum value, we need to find the minimum possible value of the quadratic expression.

The minimum value is obtained when $4x + 4 = 0$ [$d/dx = 0$]

$$x = -1.$$

The expression comes as 7.

The entire expression becomes $3/7$.

Hence, $[\frac{3}{7}, \frac{1}{2})$

VIDEO SOLUTION

57. D

Now as per the given series :

$$\text{we get } x_1 = 1 + 2 = 3$$

$$\text{Now } x_1 - x_2 = 8$$

$$\text{so } x_2 = -5$$

$$\text{Now } x_1 - x_2 + x_3 = 15$$

$$\text{so } x_3 = 7$$

$$\text{so we get } x_n = (-1)^{n+1} (2n + 1)$$

$$\text{so } x_{49} = 99 \text{ and } x_{50} = -101$$

$$\text{Therefore } x_{49} + x_{50} = -2$$

VIDEO SOLUTION

58. C

$$\text{Case 1 : } x \geq \frac{40}{3}$$

$$\text{we get } 3x - 20 + 3x - 40 = 20$$

$$6x = 80$$

$$x = \frac{80}{6} = \frac{40}{3} = 13.33$$

$$\text{Case 2: } \frac{20}{3} \leq x < \frac{40}{3}$$

we get $3x - 20 + 40 - 3x = 20$

we get $20 = 20$

So we get $x \in \left[\frac{20}{3}, \frac{40}{3} \right]$

$$\text{Case 3 } x < \frac{20}{3}$$

we get $20 - 3x + 40 - 3x = 20$

$$40 = 6x$$

$$x = \frac{20}{3}$$

but this is not possible

so we get from case 1, 2 and 3

$$\frac{20}{3} \leq x \leq \frac{40}{3}$$

Now looking at options

we can say only option C satisfies for all x .

Hence $7 < x < 12$.

[VIDEO SOLUTION](#)

59. C

Let Entire work be W

Now Anil worked for 24 days

Bimal worked for 14 days and Charu worked for 14 days.

Now Anil Completes W in 60 days

so in 24 days he completed $0.4W$

Bimal completes W in 84 Days

So in 14 Days Bimal completes $= \frac{W}{6}$

Therefore work done by charu $= W - \frac{W}{6} - \frac{4W}{10} = \frac{26W}{10} = \frac{13W}{5}$

Therefore proportion of Charu $= \frac{13}{30} \times 21000 = 9100$

[VIDEO SOLUTION](#)

60. 250

Let the number of white balls be x and black balls be y

So we get $x + y = 450$ (1)

Now metallic black balls $= 0.5y$

Metallic white balls $= 0.4x$

From condition $0.4x = 0.5y$

we get $4x - 5y = 0$ (2)

Solving (1) and (2) we get

$x = 250$ and $y = 200$

Now number of Non Metallic balls $= 0.6x + 0.5y = 150 + 100 = 250$

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61. 10

Let Total matches played be n and in initial $n-10$ matches his goals be x

so we get $\frac{(x+1)}{n} = 0.15$

we get $x + 1 = 0.15n$ (1)

From condition (2) we get :

$$\frac{(x+2)}{n} = 0.2$$

we get $x + 2 = 0.2n$ (2)

Subtracting (1) and (2)

we get $1 = 0.05n$

$n = 20$

So initially he played $n-10 = 10$ matches

[VIDEO SOLUTION](#)

62. B

Considering the three kinds of tea are A, B, and C.

The price of kind A = Rs 800 per kg.

The price of kind B = Rs 500 per kg.

The price of kind C = Rs 300 per kg.

They were mixed in the ratio of 2 : 3 : 5.

1/6 of the total mixture is sold for Rs 700 per kg.

Assuming the ratio of mixture to A = 12kg, B = 18kg, C = 30 kg.

The total cost price is $800 \times 12 + 500 \times 18 + 300 \times 30 = \text{Rs } 27600$.

Selling 1/6 which is 10kg for Rs 700/kg the revenue earned is Rs 7000.

In order to have an overall profit of 50 percent on Rs 27600.

Thes selling price of the 60 kg is Rs 27600*1.5 = Rs 41400.

Hence he must sell the remaining 50 kg mixture for Rs 41400 - Rs 7000 = 34400.

Hence the price per kg is Rs 34400/50 = Rs 688

[VIDEO SOLUTION](#)

63. D

We have :

$$x^2 - xy - x = 22 \quad (1)$$

$$\text{And } y^2 - xy + y = 34 \quad (2)$$

Adding (1) and (2)

$$\text{we get } x^2 - 2xy + y^2 - x + y = 56$$

$$\text{we get } (x - y)^2 - (x - y) = 56$$

Let (x-y) = t

$$\text{we get } t^2 - t = 56$$

$$t^2 - t - 56 = 0$$

$$(t-8)(t+7) = 0$$

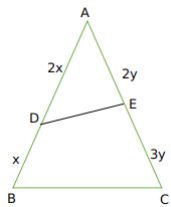
$$\text{so } t = 8$$

$$\text{so } x - y = 8$$

[VIDEO SOLUTION](#)

64. 30

We have :



$$\text{Now area of ADE} = \frac{1}{2} \times AD \times AE \times \sin A$$

$$= \frac{1}{2} \times 2x \times 2y \times \sin A = 8$$

$$\text{we get } xy \sin A = 4$$

$$\text{Now Area of triangle ABC} = \frac{1}{2} AB \times AC \times \sin A$$

$$\text{we get } \frac{1}{2} \times 3x \times 5y \sin A = \frac{15}{2} xy \sin A = \frac{15}{2} \times 4$$

$$\text{we get Area of ABC} = 30$$

[VIDEO SOLUTION](#)

65. A

Let the amount invested by Anil Bobby and Chintu be x, y, and z.

Considering $x+y+z = 100p$.

Given Anil's share was 70 percent = $70p$.

As per the information provided :

His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%.

Since the profits are distributed in the ratio of their investments :

With a 3% decrease in the profits the value of profit earned by A decreased by Rs 420 which was 70 percent of the total invested.

$$\text{Hence for all three of them would be combinedly losing } (420) \cdot \left(\frac{10}{7}\right) = 600$$

Hence 3 percent profit was equivalent to Rs 600.

The initial investment is equivalent to Rs 20000.

This is the total amount invested.

Chintu's profit share increased by Rs 80 when the profit percentage increased by 2 %. A 2 percent increase in profit is equivalent to Rs 20000*2/100 = Rs 400.

Of which Rs 80 is earned by Chintu which is 20% of the total Rs 400.

Hence he invested 20% of the total amount.

Bobby invested the other 10 percent.

10 percent of Rs 20000 = Rs 2000

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66. A

Let A fill the tank at x liters/hour and B drain it at y liters/hour

Now as per Condition 1 :

We get Volume filled till 10pm = $8x - 7y$ (1) .

Here A operates for 8 hours and B operates for 7 hours .

As per condition 2

We get Volume filled till 6pm = $4x - 2y$ (2)

Here A operates for 4 hours and B operates for 2 hours .

Now equating (1) and (2)

we get $8x - 7y = 4x - 2y$

so we get $4x = 5y$

$y = 4x/5$

So volume of tank = $8x - 7 \times \frac{4x}{5} = \frac{12x}{5}$

So time taken by A alone to fill the tank = $\frac{\frac{12x}{5}}{x} = \frac{12}{5} hrs$
= 144 minutes

 VIDEO SOLUTION

